

Computer Networks: TCP/IP

1. Layered communication

The four-layer TCP/IP model separates application, transport, internet and link responsibilities. HTTP and DNS are application protocols; TCP and UDP are transport protocols; IP handles addressing and routing; Ethernet is a link technology. Encapsulation adds protocol information as data travels down the stack.

2. TCP versus UDP

TCP provides a reliable, ordered byte stream using sequence numbers, acknowledgements and retransmissions. It also supports flow control and congestion control. TCP does not preserve application message boundaries, so applications need their own framing. UDP sends independent datagrams without built-in delivery or ordering guarantees. Applications may implement reliability above UDP.

3. Connection establishment

A typical TCP connection begins with SYN, SYN-ACK and ACK. This handshake synchronizes sequence numbers. TCP reliability does not guarantee that a disconnected application will receive data indefinitely or that an application has processed received bytes.

4. Addressing and names

An IP address identifies a network interface in a routing context. A transport port identifies an application endpoint. DNS resolves names to records, including address records. In IPv4, ARP resolves a local next-hop IPv4 address to a link-layer address; it does not resolve domain names.

5. Switching and routing

A typical Ethernet switch forwards frames using MAC addresses. A router forwards packets between IP networks. To reach a remote subnet, a host normally sends its frame to the default gateway, not directly to the remote destination MAC address.

Exam checks

Explain why TCP needs application framing. Distinguish DNS from ARP. Compare reliability, ordering and message boundaries in TCP and UDP. Avoid claiming UDP is always faster: performance depends on the application and network.

IPv4 Subnetting: Worked Example

1. Prefix and host bits

IPv4 addresses contain 32 bits. A /26 prefix uses 26 network bits and leaves 6 host bits. The mask is 255.255.255.192. Each /26 block contains $2^6 = 64$ addresses. For an ordinary /26 subnet, 62 are usable host addresses after excluding network and broadcast addresses.

2. Split a /24 into /26 subnets

Borrowing two bits from 192.168.10.0/24 creates four /26 subnets. Their network addresses are 192.168.10.0, 192.168.10.64, 192.168.10.128 and 192.168.10.192. The block size in the last octet is 64.

3. Locate a host

For 192.168.10.70/26, 70 falls in the 64-127 block. Network: 192.168.10.64. Broadcast: 192.168.10.127. Usable range: 192.168.10.65 through 192.168.10.126. Host .70 and host .100 are in the same /26; host .130 is not.

4. Choosing a subnet size

To support at least 50 ordinary host interfaces, choose h with $2^h - 2$ at least 50. Six host bits give 62 usable addresses, so /26 is sufficient. Five host bits give only 30, so /27 is insufficient.

5. Important exceptions

The subtract-two rule applies to conventional subnets, not every prefix. A /31 can use both addresses on a point-to-point link, and /32 denotes one address. Do not apply the /26 host rule mechanically to those cases.

Practice with answer check

For 192.168.10.150/26, find network, broadcast and usable range. Answer: network .128, broadcast .191, usable .129-.190. Explain why .150 and .70 are in different subnets even though their first three octets match.